

Introduction to Data Assimilation

Lecture 2 Supplementary

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1 Setting and notation

We consider the Itô SDE on $[0, T]$

$$dX_t = A(X_t, t) dt + B(X_t, t) dW_t, \quad X_0 \in L^2, \quad (1.1)$$

where W_t is a standard m -dimensional Wiener process and $X_t \in \mathbb{R}^d$. We assume global Lipschitz and linear growth:

$$\|A(x, t) - A(y, t)\| + \|B(x, t) - B(y, t)\| \leq L\|x - y\|, \quad (1.2)$$

$$x^\top A(x, t) + \frac{1}{2}\|B(x, t)\|_F^2 \leq C(1 + \|x\|^2). \quad (1.3)$$

Let $0 = t_0 < t_1 < \dots < t_N = T$ be a uniform grid with $\Delta t = t_{n+1} - t_n$ and $\Delta W_n := W_{t_{n+1}} - W_{t_n}$.

2 Generating Wiener increments in practice

Definition 2.1 (Scalar increments). For scalar Brownian motion, the increments are independent Gaussians:

$$\Delta W_n \sim \mathcal{N}(0, \Delta t), \quad \text{generate as} \quad \Delta W_n = \sqrt{\Delta t} Z_n, \quad Z_n \sim \mathcal{N}(0, 1).$$

Definition 2.2 (Multi-dimensional increments). For m -dimensional W_t , take $Z_n \sim \mathcal{N}(0, I_m)$ and set $\Delta \mathbf{W}_n = \sqrt{\Delta t} Z_n$ (independent components). If a covariance $Q \in \mathbb{R}^{m \times m}$ is prescribed (correlated Wiener),

$$\Delta \mathbf{W}_n \sim \mathcal{N}(0, \Delta t Q), \quad \Delta \mathbf{W}_n = \sqrt{\Delta t} L Z_n, \quad \text{with } Q = LL^\top \text{ (Cholesky)}.$$

Remark 2.3 (Random seeds). Fix the RNG seed when comparing schemes to ensure identical noise realizations; this isolates the *discretization* error from sampling variability.

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3 Euler–Maruyama method and convergence

Definition 3.1 (Euler–Maruyama (EM)). The EM update for (1.1) is

$$X_{n+1} = X_n + A(X_n, t_n) \Delta t + B(X_n, t_n) \Delta W_n. \quad (3.1)$$

Definition 3.2 (Strong and weak errors). Let $X(t_n)$ be the exact solution. The *strong error* at t_n is $e_n^{\text{str}} := (\mathbb{E} \|X_n - X(t_n)\|^2)^{1/2}$. The scheme is *strong order* p if $\max_n e_n^{\text{str}} = O(\Delta t^p)$. For smooth test φ ,

$$e_n^{\text{weak}} := |\mathbb{E} \varphi(X_n) - \mathbb{E} \varphi(X(t_n))|, \quad \text{weak order } p \iff \max_n e_n^{\text{weak}} = O(\Delta t^p).$$

Theorem 3.3 (Moment bounds). Under (1.2)–(1.3), $\sup_{t \in [0, T]} \mathbb{E} \|X(t)\|^2 < \infty$ and for EM, $\max_n \mathbb{E} \|X_n\|^2 \leq C$ uniformly in Δt (for small enough Δt).

Theorem 3.4 (Strong order of EM). Under (1.2)–(1.3), EM has strong order $1/2$:

$$\max_{0 \leq n \leq N} (\mathbb{E} \|X_n - X(t_n)\|^2)^{1/2} \leq C \Delta t^{1/2}.$$

Proof. Write the exact variation-of-constants over $[t_n, t_{n+1}]$:

$$X(t_{n+1}) = X(t_n) + \underbrace{\int_{t_n}^{t_{n+1}} A(X_s, s) ds}_{D_n} + \underbrace{\int_{t_n}^{t_{n+1}} B(X_s, s) dW_s}_{M_n}.$$

Subtract EM (3.1) and define $E_n := X(t_n) - X_n$:

$$E_{n+1} = E_n + (D_n - A(X_n, t_n) \Delta t) + (M_n - B(X_n, t_n) \Delta W_n).$$

Decompose $D_n = \int_{t_n}^{t_{n+1}} [A(X_s, s) - A(X_n, t_n)] ds + A(X_n, t_n) \Delta t$, so the $A(X_n, t_n) \Delta t$ cancels, and similarly for M_n . Hence

$$E_{n+1} = E_n + R_n^D + R_n^M,$$

with

$$R_n^D := \int_{t_n}^{t_{n+1}} (A(X_s, s) - A(X_n, t_n)) ds, \quad R_n^M := \int_{t_n}^{t_{n+1}} (B(X_s, s) - B(X_n, t_n)) dW_s.$$

Take squared norms and expectations, use $(a + b + c)^2 \leq (1 + \eta)a^2 + C_\eta(b^2 + c^2)$ for any $\eta > 0$:

$$\mathbb{E} \|E_{n+1}\|^2 \leq (1 + \eta) \mathbb{E} \|E_n\|^2 + C_\eta (\mathbb{E} \|R_n^D\|^2 + \mathbb{E} \|R_n^M\|^2).$$

By Lipschitz and Jensen,

$$\|A(X_s, s) - A(X_n, t_n)\| \leq L \|X_s - X_n\| + L |s - t_n|.$$

Using moment bounds (Thm. 3.3) and standard Hölder continuity of X_t in L^2 ($\mathbb{E}\|X_s - X_{t_n}\|^2 \leq C|s - t_n|$), one gets

$$\mathbb{E}\|R_n^D\|^2 \leq C(\Delta t)^2.$$

For the martingale remainder, BDG inequality with Lipschitz gives

$$\mathbb{E}\|R_n^M\|^2 \leq C \mathbb{E} \int_{t_n}^{t_{n+1}} \|B(X_s, s) - B(X_n, t_n)\|^2 ds \leq C(\Delta t)^2 + C \Delta t \mathbb{E}\|E_n\|^2.$$

Combine:

$$\mathbb{E}\|E_{n+1}\|^2 \leq (1 + C\Delta t) \mathbb{E}\|E_n\|^2 + C(\Delta t)^2.$$

Discrete Grönwall yields $\mathbb{E}\|E_n\|^2 \leq C \Delta t$, i.e. strong order 1/2. $\square$

Theorem 3.5 (Weak order of EM). *If A, B and test φ are sufficiently smooth with polynomial growth, then EM has weak order 1:*

$$\max_n |\mathbb{E} \varphi(X_n) - \mathbb{E} \varphi(X(t_n))| \leq C \Delta t.$$

Idea (Talay–Tubaro expansion). Let $u(x, t) = \mathbb{E}[\varphi(X_T) \mid X_t = x]$, so u solves the backward Kolmogorov PDE

$$\partial_t u + \mathcal{L}u = 0, \quad \mathcal{L}\psi = A \cdot \nabla \psi + \frac{1}{2}(BB^\top) : \nabla^2 \psi.$$

Compute one-step weak error

$$\epsilon_n := |\mathbb{E} u(X_{n+1}, t_{n+1}) - \mathbb{E} u(X(t_{n+1}), t_{n+1})|.$$

A second-order Itô–Taylor expansion of $u(X_{n+1}, t_{n+1})$ around (X_n, t_n) and matching with the generator gives $\epsilon_n \leq C(\Delta t)^2$ uniformly. Summation over $N = T/\Delta t$ steps yields $O(\Delta t)$. $\square$

3.1 Mathematical Tools

Theorem 3.6 (Burkholder–Davis–Gundy (BDG) Inequality). *Let $(M_t)_{t \geq 0}$ be a continuous local martingale with $M_0 = 0$ and quadratic variation process*

$$\langle M \rangle_t = \int_0^t |H_s|^2 ds,$$

where H_s is a predictable (adapted) process. Then for any $p \geq 1$, there exist constants $C_p, C'_p > 0$ such that

$$C_p^{-1} \mathbb{E}[\langle M \rangle_T^{p/2}] \leq \mathbb{E}\left[\sup_{0 \leq t \leq T} |M_t|^p\right] \leq C'_p \mathbb{E}[\langle M \rangle_T^{p/2}]. \quad (3.2)$$

In particular, for stochastic integrals of the form

$$M_t = \int_0^t H_s dW_s,$$

we have

$$\mathbb{E} \left[\sup_{0 \leq t \leq T} \left| \int_0^t H_s dW_s \right|^p \right] \leq C_p \mathbb{E} \left[\left(\int_0^T |H_s|^2 ds \right)^{p/2} \right].$$

Proof idea. The result is nontrivial and relies on martingale inequalities and stopping-time arguments. The intuition is that the L^p -norm of the maximal fluctuation of a martingale M_t is equivalent (up to constants) to the L^p -norm of its quadratic variation $\langle M \rangle_T^{1/2}$, i.e. the total accumulated variance. $\square$

Remark 3.7 (Practical use in SDE analysis). In SDE convergence proofs, the BDG inequality is typically applied to stochastic remainder terms such as

$$R_n^M = \int_{t_n}^{t_{n+1}} (B(X_s, s) - B(X_n, t_n)) dW_s.$$

Taking expectations and applying BDG with $p = 2$ gives

$$\mathbb{E} \|R_n^M\|^2 \leq C \mathbb{E} \left[\int_{t_n}^{t_{n+1}} \|B(X_s, s) - B(X_n, t_n)\|^2 ds \right].$$

This converts the difficult stochastic integral into a deterministic time integral, which can then be bounded using Lipschitz continuity and moment estimates of X_t .

Remark 3.8 (Historical note). The inequality is named after Donald Burkholder, Benjamin Davis, and Ronald Gundy, who established it in the 1960s. It's a beautiful generalization of Doob's maximal inequality, connecting martingale oscillations with their "energy." In stochastic numerics, BDG is one of the main tools to control $\mathbb{E} \|X_t - X_n\|^p$.

Theorem 3.9 (Discrete Grönwall Inequality). *Let $\{a_n\}, \{b_n\}, \{c_n\}$ be nonnegative sequences satisfying*

$$a_n \leq b_n + \sum_{k=0}^{n-1} c_k a_k, \quad n \geq 1. \quad (3.3)$$

If $c_k \leq C$ for all k , then

$$a_n \leq b_n + C \sum_{k=0}^{n-1} b_k e^{C(n-k-1)}. \quad (3.4)$$

In particular, if $b_n = C_1 \Delta t$ and $c_n = C_2 \Delta t$, then

$$a_n \leq C_1 \Delta t \sum_{k=0}^{n-1} (1 + C_2 \Delta t)^{n-k-1} \leq \frac{C_1}{C_2} (e^{C_2 n \Delta t} - 1),$$

so $a_n = O(\Delta t)$ uniformly in n for bounded $T = n \Delta t$.

Proof idea. The proof mirrors the continuous Grönwall inequality. Starting from (3.3), define $d_n = \sum_{k=0}^{n-1} c_k$. Then by induction:

$$a_n \leq b_n + \sum_{k=0}^{n-1} b_k \prod_{j=k+1}^{n-1} (1 + c_j).$$

Using the elementary bound $(1 + c_j) \leq e^{c_j}$ gives

$$a_n \leq b_n + \sum_{k=0}^{n-1} b_k e^{\sum_{j=k+1}^{n-1} c_j} \leq b_n + e^{d_n} \sum_{k=0}^{n-1} b_k,$$

and if $c_j \leq C$, then $e^{d_n} \leq e^{Cn}$, yielding (3.4). □

Remark 3.10 (Intuitive meaning). The inequality bounds any recursively growing quantity a_n that depends linearly on its previous values and a known source term b_n . It prevents exponential blow-up provided c_n is controlled. In convergence proofs (e.g. Euler–Maruyama), it ensures that local errors of size $O(\Delta t^p)$ accumulate only linearly, leading to global errors of order $O(\Delta t^{p-1})$ or $O(\Delta t^p)$, depending on context.